

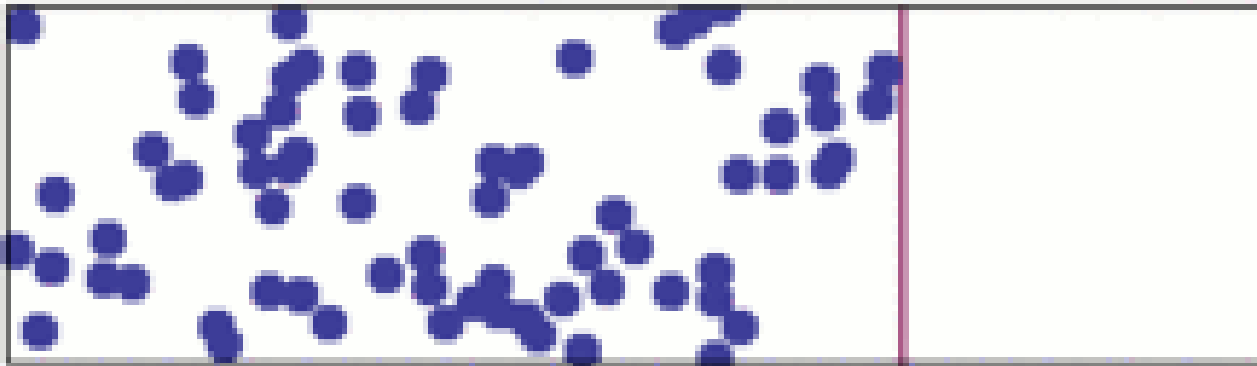
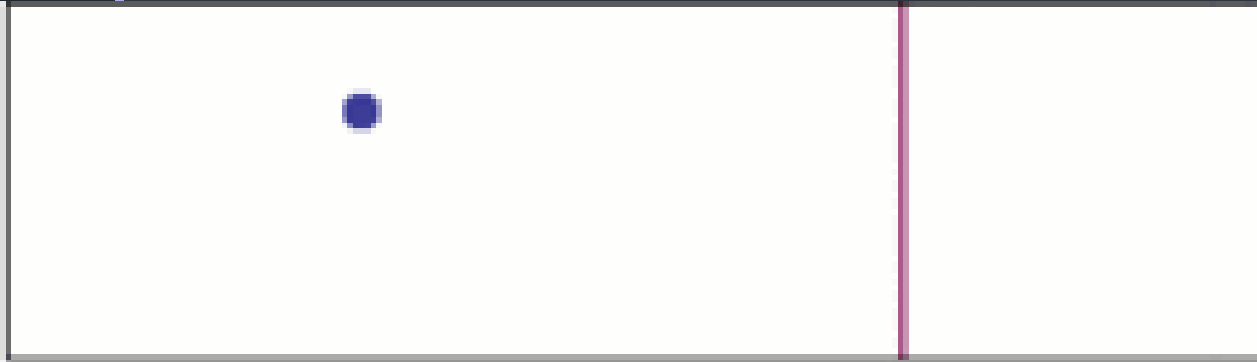
316-221: Metallurgy

Chapter :4

Diffusion

in

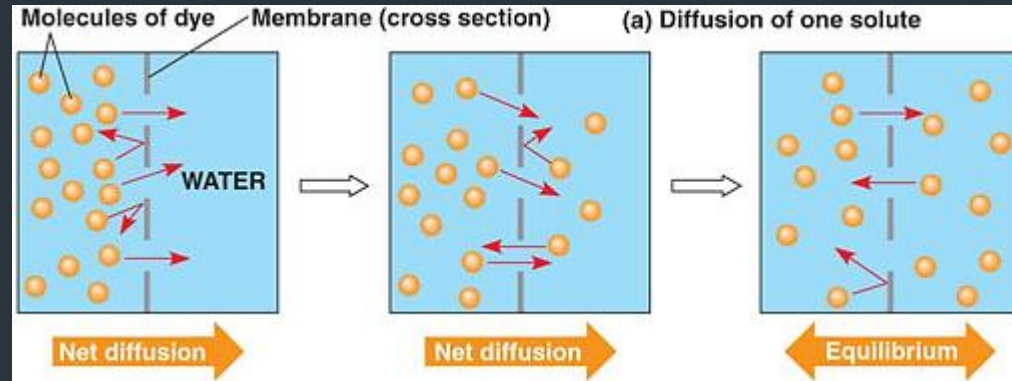
solid



Chapter 4: Diffusion in Solids

Diffusion

Diffusion – Mass transport by atomic motion



Mechanisms

- **Gases & Liquids** – random motion (Brownian motion)
- **Solids** – vacancy diffusion or interstitial diffusion

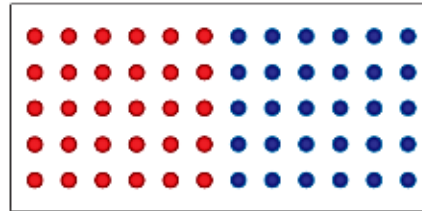
Chapter 4: Diffusion in Solids

Diffusion

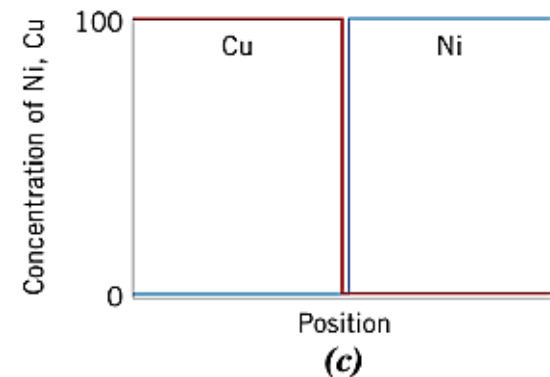
The movement of particles in a solid from an area of **high** concentration to an area of **low** concentration.



(a)



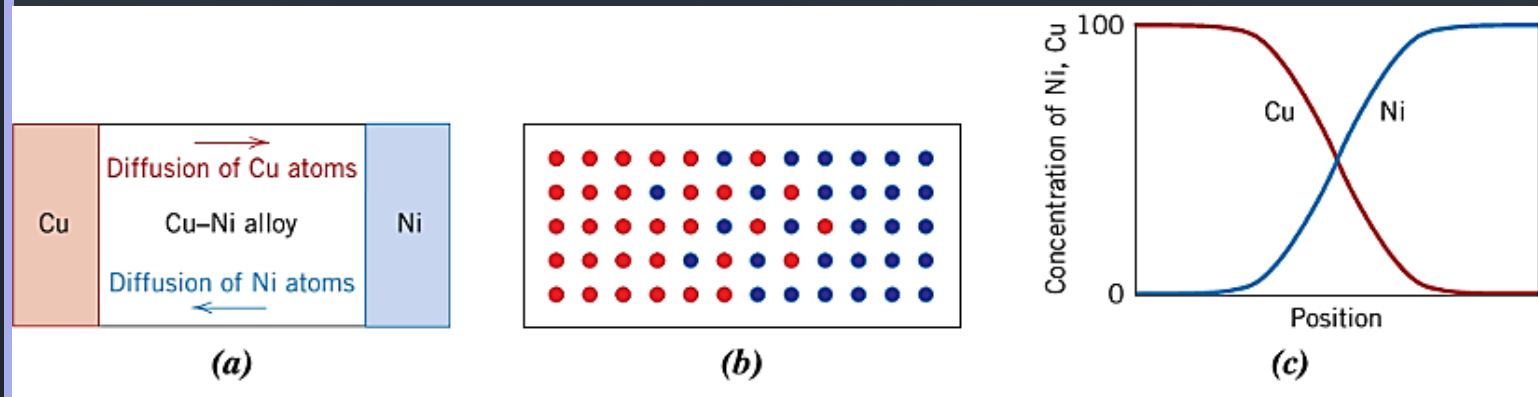
(b)



- (a) Cu-Ni diffusion couple before a high-temperature heat treatment.
- (b) Cu (**red**) and Ni (**blue**) atom locations within the diffusion couple.
- (c) Concentrations of Cu and Ni as a function of position across the couple.

Chapter 4: Diffusion in Solids

Diffusion

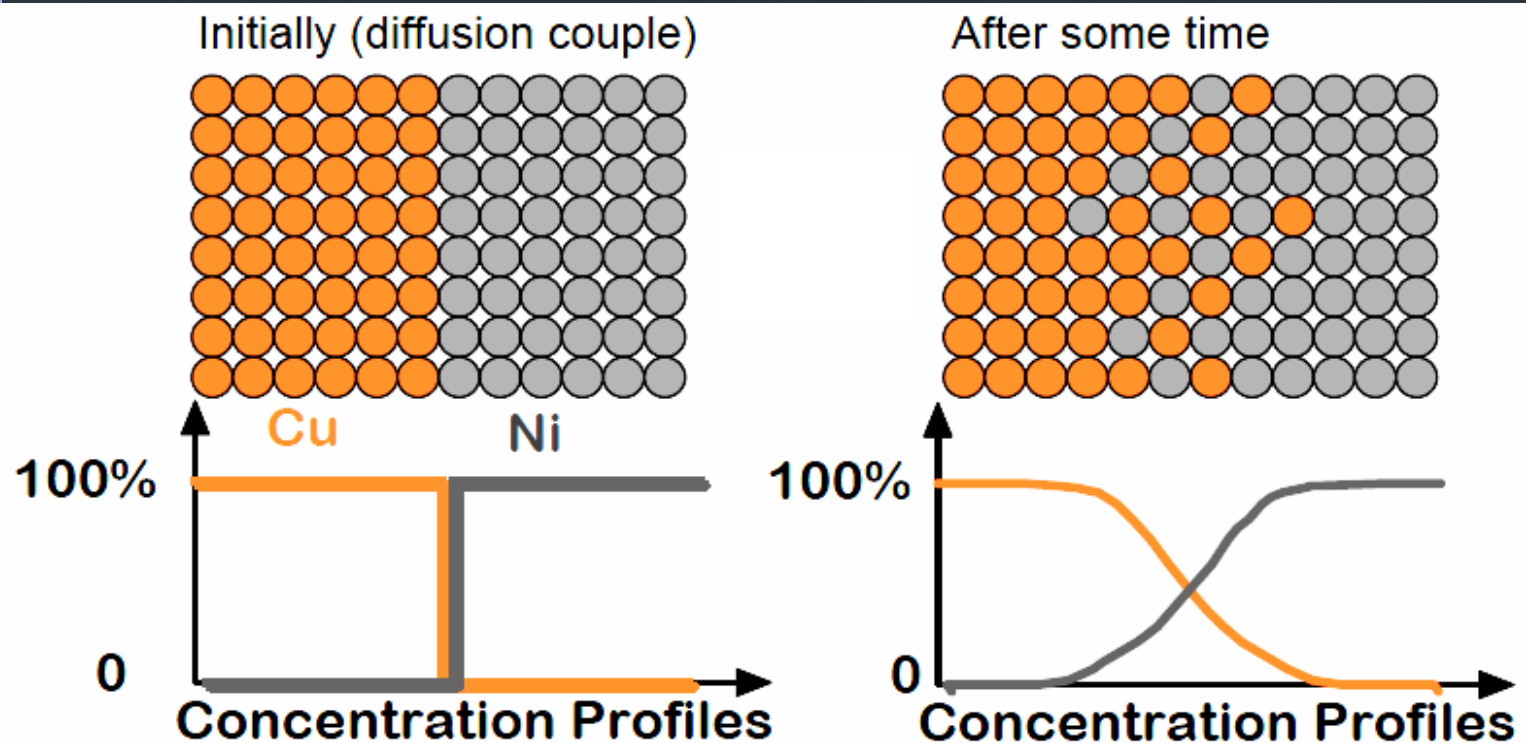


- (a) **Cu-Ni** diffusion couple after a high-temperature heat treatment, showing the alloyed diffusion zone.
- (b) Cu (**red**) and Ni (**blue**) atom locations within the couple.
- (c) Concentrations of Cu and Ni as a function of position across the couple.

Chapter 4: Diffusion in Solids

Diffusion

Interdiffusion: In an alloy or “**diffusion couple**”, atoms tend to migrate from regions of large to lower concentration.

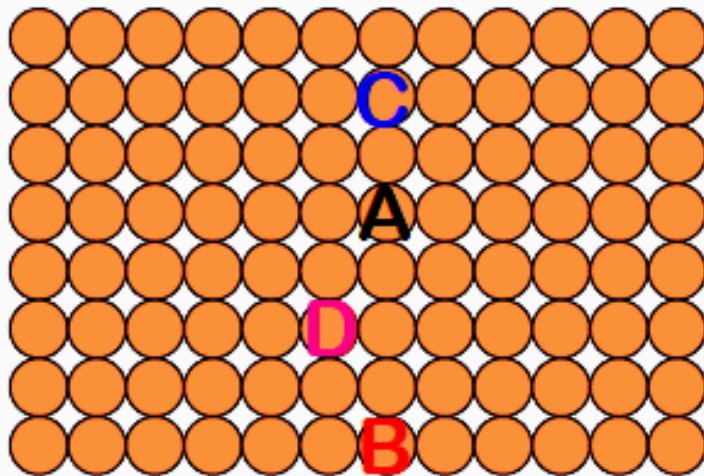


Chapter 4: Diffusion in Solids

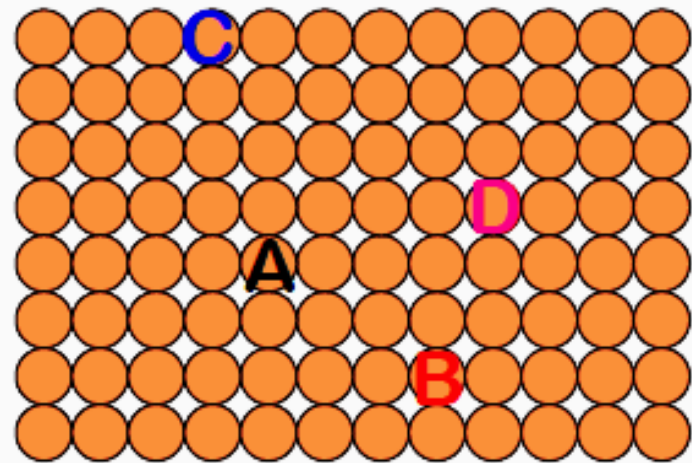
Diffusion

Self-diffusion: In an elemental solid, atoms also migrate.

Label some atoms



After some time



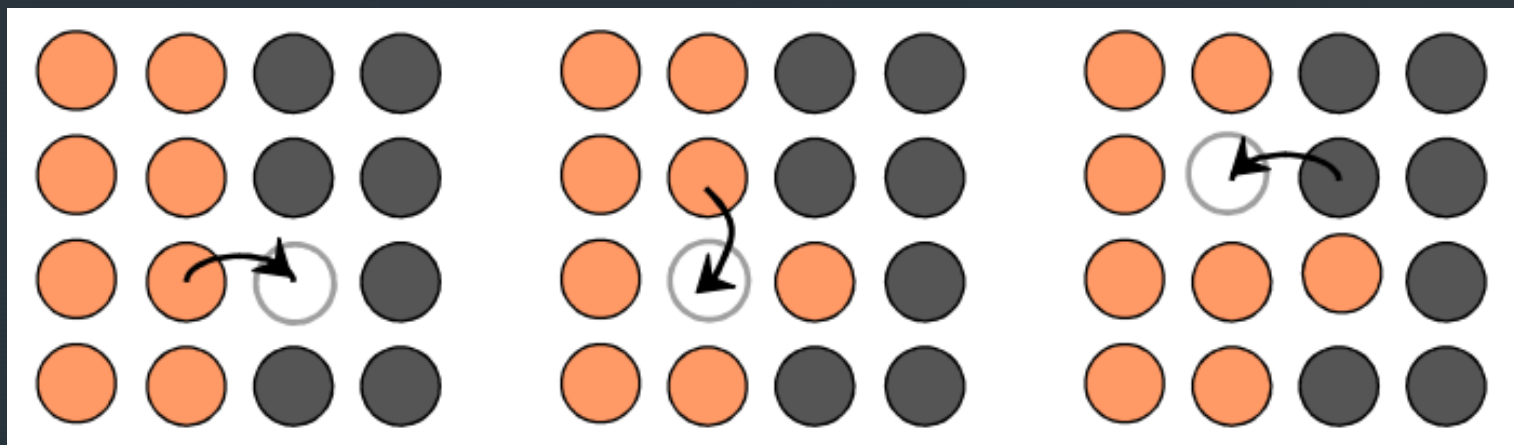
The atom must have sufficient energy to break bonds with its neighbor atoms and then cause some lattice distortion during the displacement.

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Diffusion Mechanism

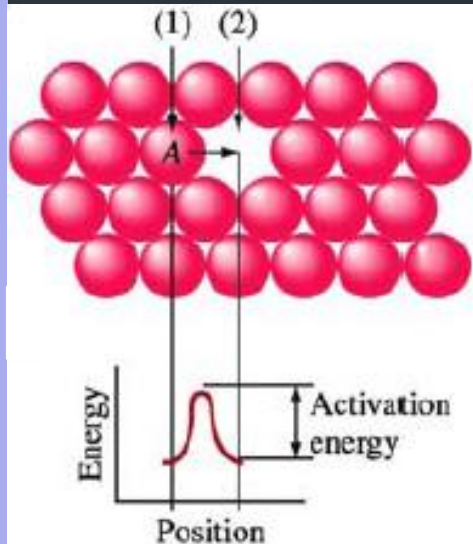
□ **Vacancy (substitutional) diffusion:** migration of atom in a lattice assisted by the presence of vacancies.

- Atoms exchange with vacancies
- Applies to substitutional impurities atoms
- Rate depends on:
 - number of vacancies
 - activation energy to exchange.



Chapter 4: Diffusion in Solids

Diffusion Mechanism



□ Higher T_{melt} → stronger bonding between atoms.

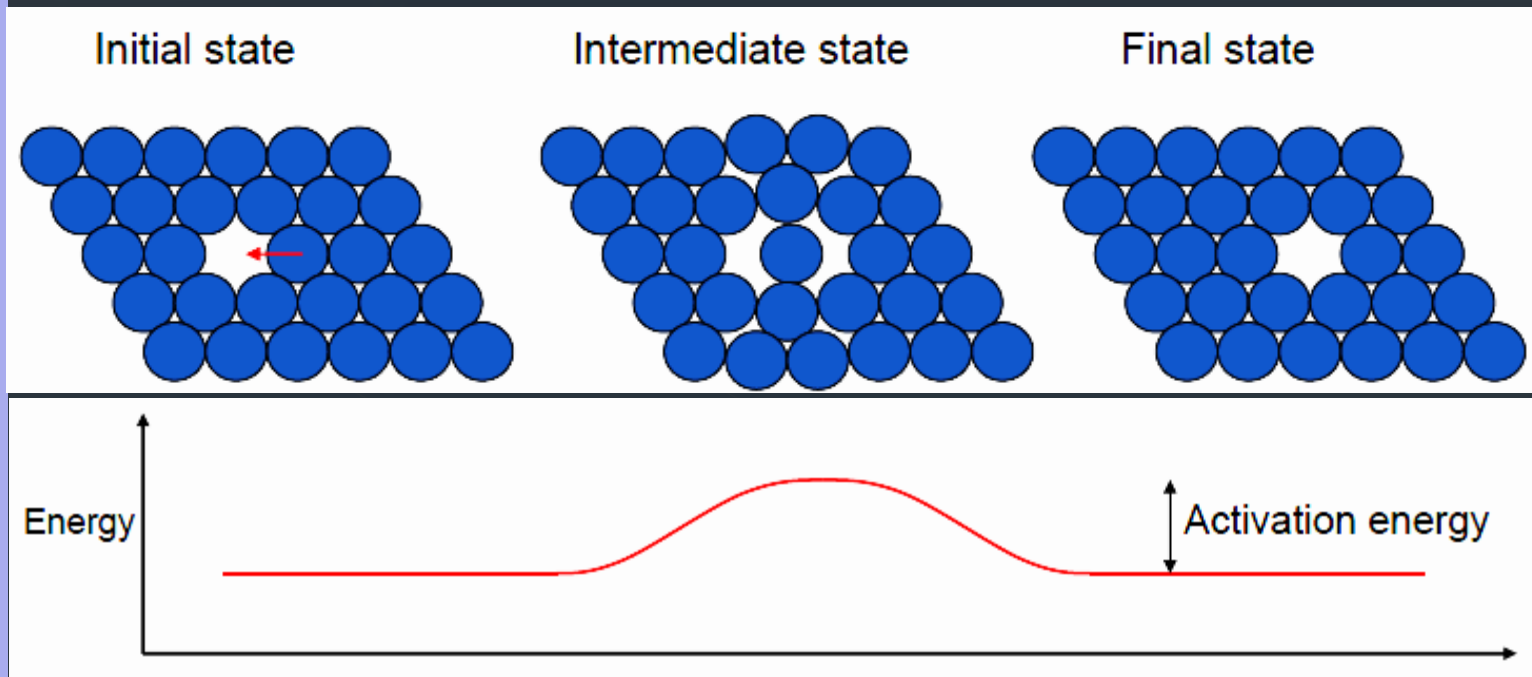
→ high activation energy to move atoms.

Metal	Melting point (°C)	Crystal structure	Temperature range studied (°C)	Activation energy	
				kJ/mol	kcal/mol
Zinc	419	HCP	240–418	91.6	21.9
Aluminum	660	FCC	400–610	165	39.5
Copper	1083	FCC	700–990	196	46.9
Nickel	1452	FCC	900–1200	293	70.1
α iron	1530	BCC	808–884	240	57.5
Molybdenum	2600	BCC	2155–2540	460	110

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Diffusion Mechanism

Activation energy for diffusion

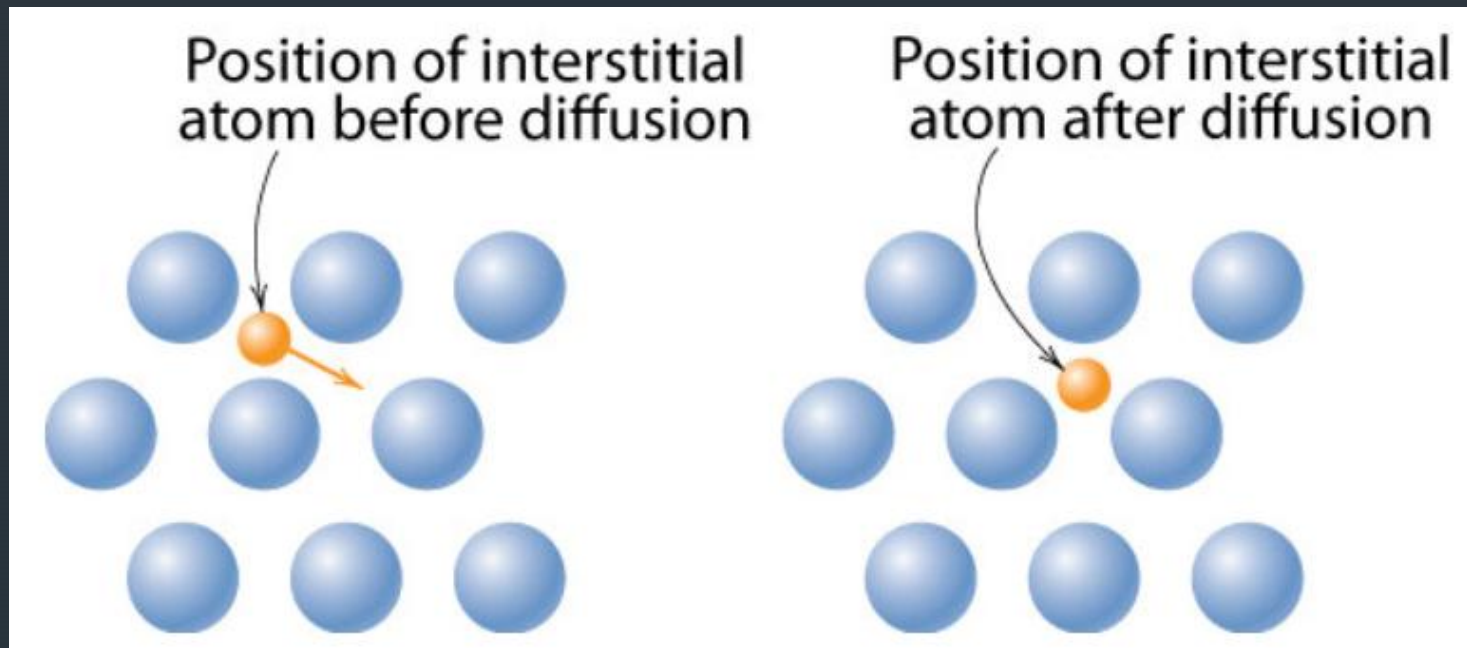


Also called energy barrier for diffusion

Chapter 4: Diffusion in Solids

Diffusion Mechanism

- ❑ **Interstitial diffusion:** smaller atoms can diffuse between atoms.



More rapid than vacancy diffusion

Chapter 4: Diffusion in Solids

Processing using diffusion



Case Hardening:

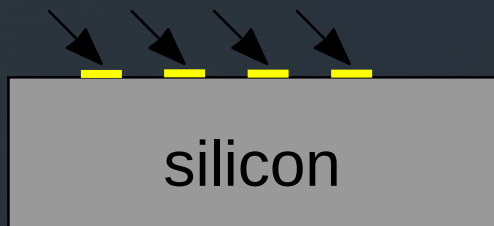
- Diffuse carbon atoms into the host iron atoms at the surface.
- Example of interstitial diffusion is a case hardened gear.
- The "case" appears as the dark outer rim of that segment of the gear that has been sectioned.

Chapter 4: Diffusion in Solids

Processing using diffusion

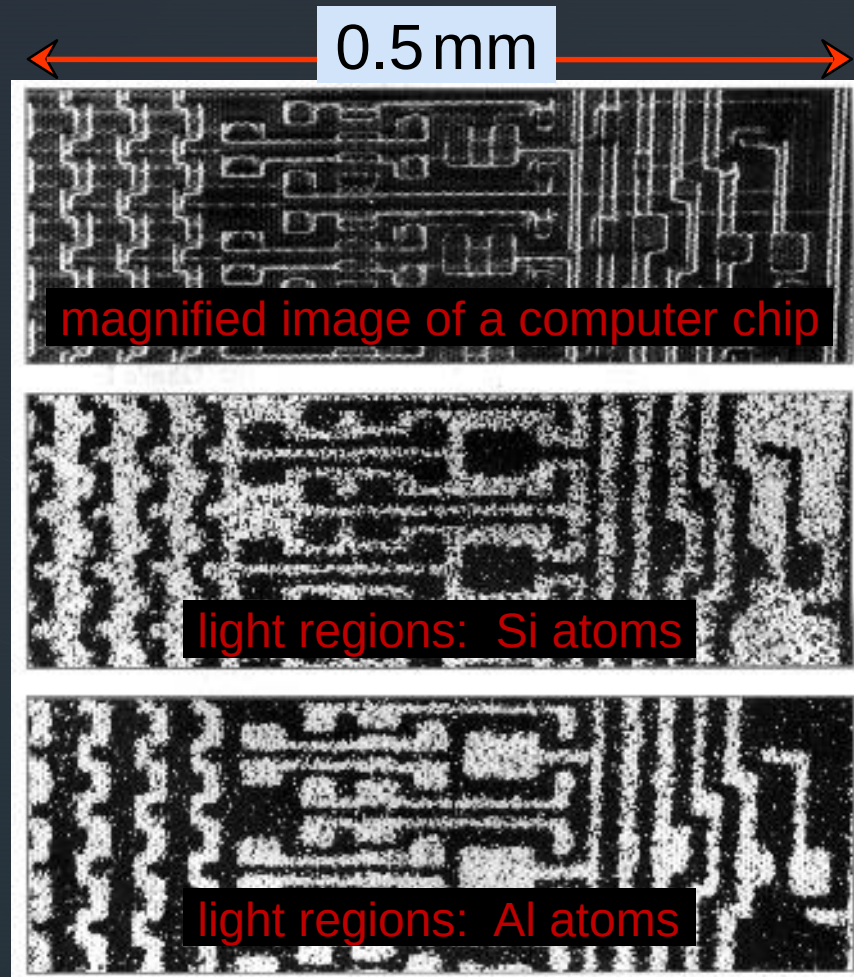
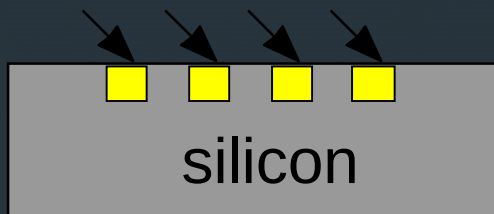
- Doping Silicon with P for n-type semiconductors:
- Process:

1. Deposit **P** rich layers on surface.



2. Heat.

3. Result: Doped semiconductor regions.



Chapter 4: Diffusion in Solids

Diffusion rate

- How do we quantify the amount or rate of diffusion?

Diffusion rate in a system will increase with temperature:

$$Rate = Ce^{-Q/RT}$$

C is proportionality constant

Q is activation energy for diffusing species, J/mol

R is molar gas constant (8.314 J mol⁻¹ K⁻¹; or 1.987 cal mol⁻¹ K⁻¹)

T is absolute temperature (K)

Chapter 4: Diffusion in Solids

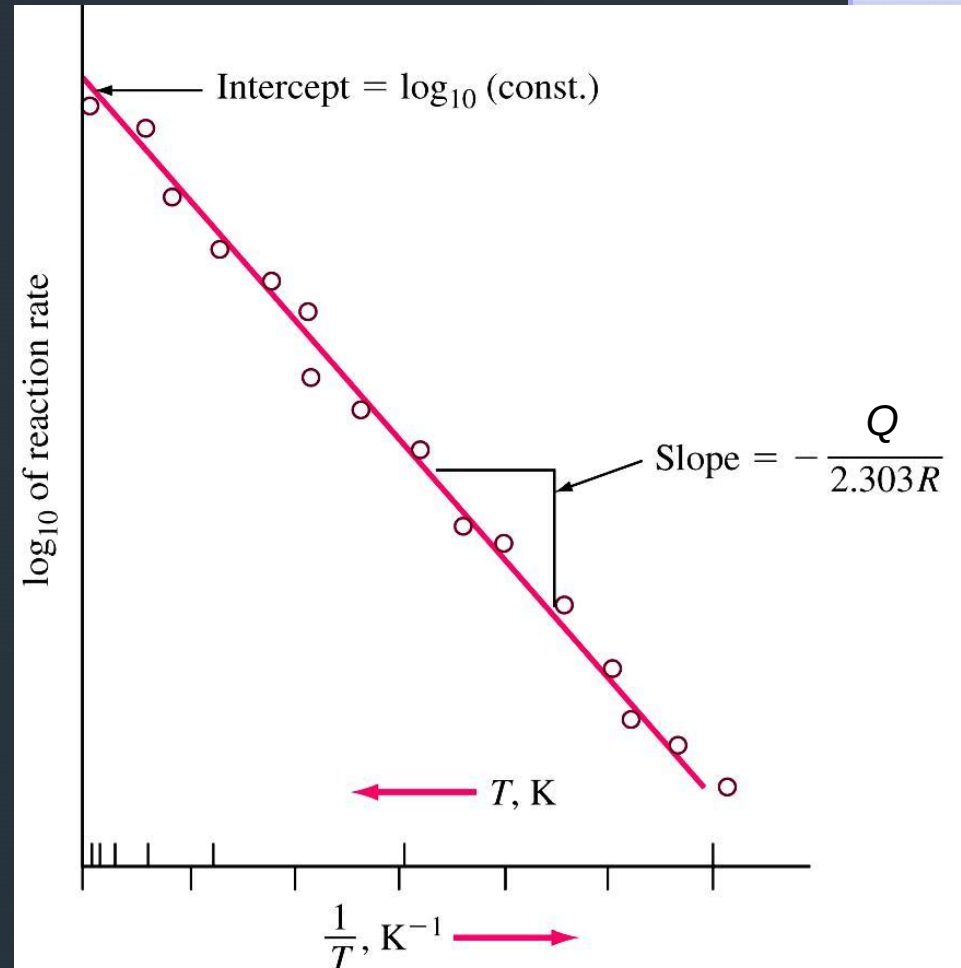
Thermal activation energy >> Typical Arrhenius Plot

>> If rewrite in natural log plot:

$$\ln rate = \ln C - \frac{Q}{RT}$$

>> or logarithmic log plot:

$$\log_{10} rate = \log_{10} C - \frac{Q}{2.303 RT}$$



Chapter 4: Diffusion in Solids

Diffusion Flux

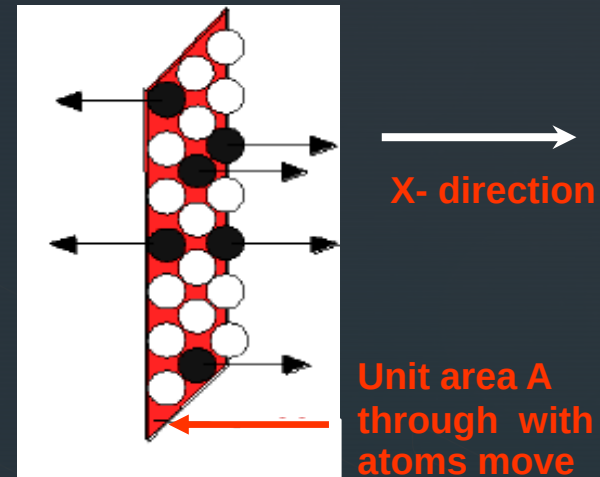
- **Flux (J):** amount of material or atoms moving past a unit area in unit time

$$J \equiv \text{Flux} \equiv \frac{\text{moles (or mass) diffusing}}{(\text{surface area})(\text{time})} = \frac{\text{mol}}{\text{cm}^2\text{s}} \text{ or } \frac{\text{kg}}{\text{m}^2\text{s}}$$

$$J = \frac{1}{A} \frac{dM}{dt} \Rightarrow \left[\frac{\text{kg}}{\text{m}^2\text{s}} \right]$$

Flux can be measured for:

- vacancies
- host (A) atoms
- impurity (B) atoms

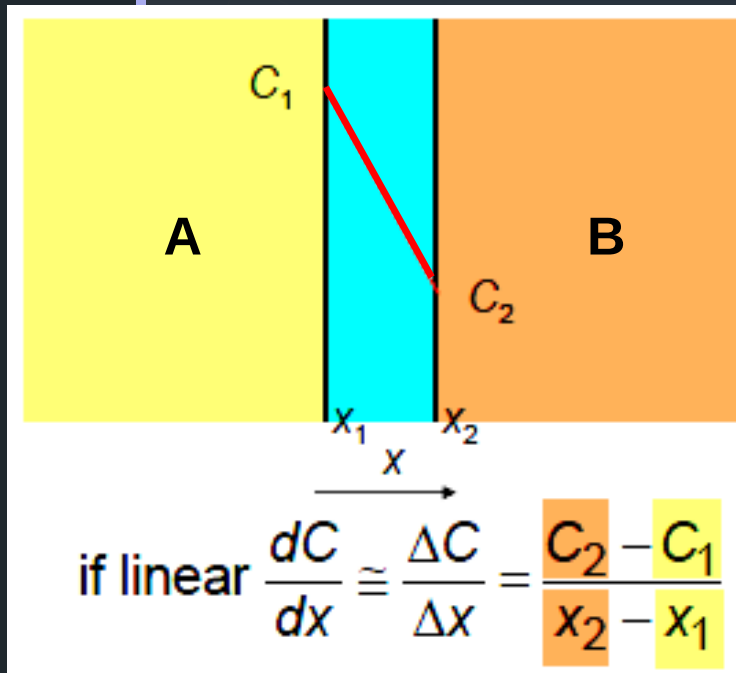


If the **diffusion flux** does not change with time, a **steady-state** condition exists.

Chapter 4: Diffusion in Solids

Steady-State Diffusion

- Rate of diffusion independent of **time (t)**
- Flux proportional to concentration gradient = $\frac{dC}{dX}$



The mathematics of **steady-state diffusion** in a single (**x**) direction

Fick's first law of diffusion

$$J = -D \frac{dC}{dX}$$

J = flux (atom/m².s)

D = diffusion coefficient (m²/s)

Chapter 4: Diffusion in Solids

Steady-State Diffusion

A plate of iron is exposed to a carburizing (carbon-rich) atmosphere on one side and a decarburizing (carbon-deficient) atmosphere on the other side at **700°C (1300°F)**. **If a condition of steady state is achieved, calculate the diffusion flux of carbon** through the plate if the concentrations of carbon at positions of **5 and 10 mm (5×10^{-3} and 10^{-2} m)** beneath the carburizing surface are **1.2 and 0.8 kg/m³**, respectively. Assume a diffusion coefficient of **3×10^{-11} m²/s** at this temperature.

Fick's first law equation;

$$J = -D \frac{dC}{dX}$$

$$\begin{aligned} J &= -D \frac{C_A - C_B}{x_A - x_B} = -(3 \times 10^{-11} \text{ m}^2/\text{s}) \frac{(1.2 - 0.8) \text{ kg/m}^3}{(5 \times 10^{-3} - 10^{-2}) \text{ m}} \\ &= 2.4 \times 10^{-9} \text{ kg/m}^2\text{-s} \end{aligned}$$

Chapter 4: Diffusion in Solids

Nonsteady-State Diffusion

The diffusion flux and the concentration gradient at some particular point in a solid **vary with time**.

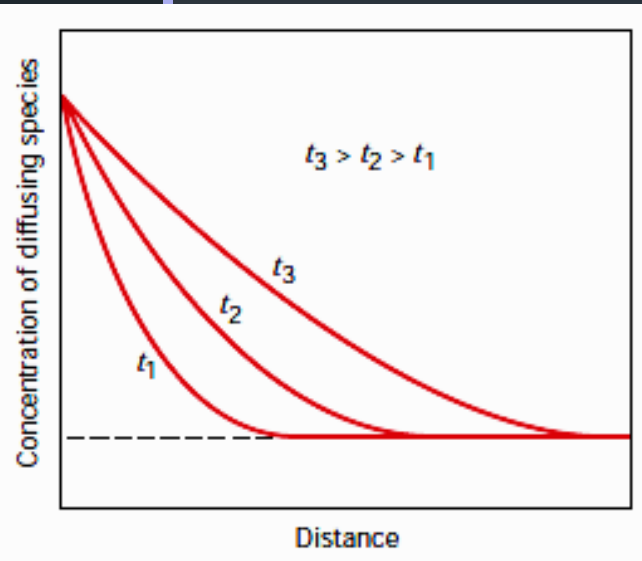
In this case **Fick's Second Law** is used:

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right)$$

If the diffusion coefficient is independent of composition:

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

The concentration profiles at three different diffusion times under conditions of **nonsteady state**.



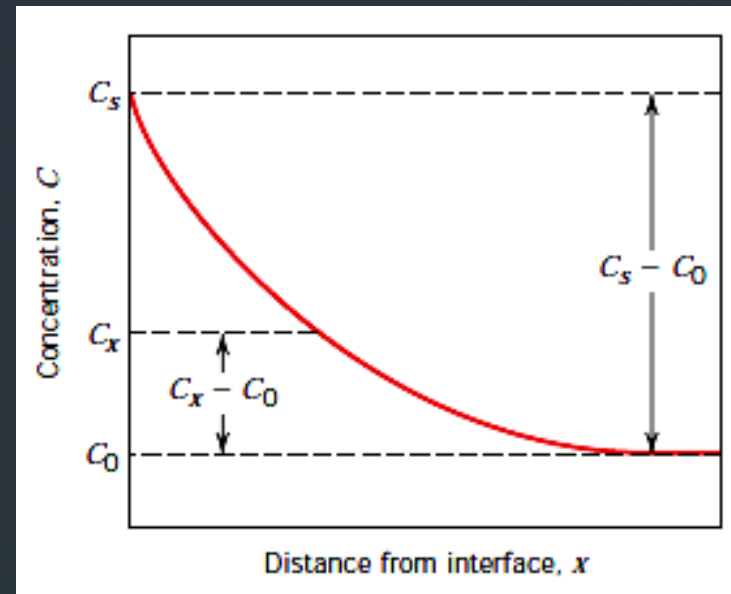
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Nonsteady-State Diffusion

The solution to this differential equation with the given boundary condition:

$$\frac{C_x - C_o}{C_s - C_o} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

- C_s is surface concentration
- C_o is initial concentration
- C_x is concentration at distance x from surface
- x is distance from surface
- D is diffusion coefficient
- t is time
- erf is error function



Chapter 4: Diffusion in Solids

Nonsteady-State Diffusion

Table 6.1 Tabulation of Error Function Values

z	$\text{erf}(z)$	z	$\text{erf}(z)$	z	$\text{erf}(z)$
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

Note: z is the independent variable, i.e.
and $\text{erf}(z)$ is the dependent variable.

$$Z = \frac{x}{2\sqrt{Dt}}$$

Chapter 4: Diffusion in Solids

Nonsteady-State Diffusion

For some applications, it is necessary to **harden the surface** of a steel (or iron-carbon alloy) above that of its interior. One way this may be accomplished is by increasing the surface concentration of carbon in a process termed **carburizing**; the steel piece is exposed, at an elevated temperature, to an atmosphere rich in a hydrocarbon gas, such as methane (CH_4).

Consider one such alloy that initially has a uniform carbon concentration of **0.25 wt%** and is to be treated at **950°C (1750°F)**. If the concentration of carbon at the surface is suddenly brought to and maintained at **1.20 wt%**, how long will it take to achieve a carbon content of **0.80 wt%** at a position **0.5 mm** below the surface? The diffusion coefficient for carbon in iron at this temperature is **$1.6 \times 10^{-11} \text{ m}^2/\text{s}$** ; assume that the steel piece is semi-infinite.

Chapter 4: Diffusion in Solids

Nonsteady-State Diffusion

Solution

Since this is a nonsteady-state diffusion problem in which the surface composition is held constant. Values for all the parameters in this expression except time (t) are specified in the problem as follows:

$$C_0 = 0.25 \text{ wt\% C}$$

$$C_s = 1.20 \text{ wt\% C}$$

$$C_x = 0.80 \text{ wt\% C}$$

$$x = 0.50 \text{ mm or } 5 \times 10^{-4} \text{ m}$$

$$D = 1.6 \times 10^{-11} \text{ m}^2/\text{s}$$

Thus,

$$\frac{C_x - C_0}{C_s - C_0} = \frac{0.80 - 0.25}{1.20 - 0.25} = 1 - \operatorname{erf} \left[\frac{(5 \times 10^{-4} \text{ m})}{2\sqrt{(1.6 \times 10^{-11} \text{ m}^2/\text{s}) (t)}} \right]$$

$$0.4210 = \operatorname{erf} \left(\frac{62.5 \text{ s}^{1/2}}{\sqrt{t}} \right)$$

Chapter 4: Diffusion in Solids

Nonsteady-State Diffusion

We must now determine from **erf table**, the value of **z** for which the **erf is 0.4210**. An interpolation is necessary, as

z	$\text{erf}(z)$
0.35	0.3794
z	0.4210
0.40	0.4284

$$\frac{z - 0.35}{0.40 - 0.35} = \frac{0.4210 - 0.3794}{0.4284 - 0.3794}$$

$$z = 0.392$$

Therefore,

$$\frac{62.5 \text{ s}^{1/2}}{\sqrt{t}} = 0.392$$

And solving for **t**,

$$t = \left(\frac{62.5 \text{ s}^{1/2}}{0.392} \right)^2 = 25,400 \text{ s} = 7.1 \text{ h}$$

Chapter 4: Diffusion in Solids

Factors that Influence Diffusion

□ Diffusing species

- magnitude of diffusion coefficient, (D)
 - indicates the rate at which atoms diffuse
- both diffusing species and host material influence the coefficient
 - Relative sizes of atoms
 - “Openness” of lattice
 - Ionic charges

Chapter 4: Diffusion in Solids

Factors that Influence Diffusion

For example: Atomic size

- For the host species of iron:
 - Self diffusion at 500°C (**Fe moving in Fe**)
 $D = 1.1 \times 10^{-20} \text{ m}^2/\text{s}$ (**vacancy diffusion**)
 - Carbon interdiffusion at 500°C (**C moving in Fe**)
 $D = 2.3 \times 10^{-12} \text{ m}^2/\text{s}$ (**interstitial diffusion**)

This shows the contrast between rates of vacancy and interstitial diffusion.

Chapter 4: Diffusion in Solids

Factors that Influence Diffusion

□ Temperature

- Diffusion coefficient increases with increasing T

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

D is diffusion coefficient (m^2/s)

D_0 is pre-exponential constant

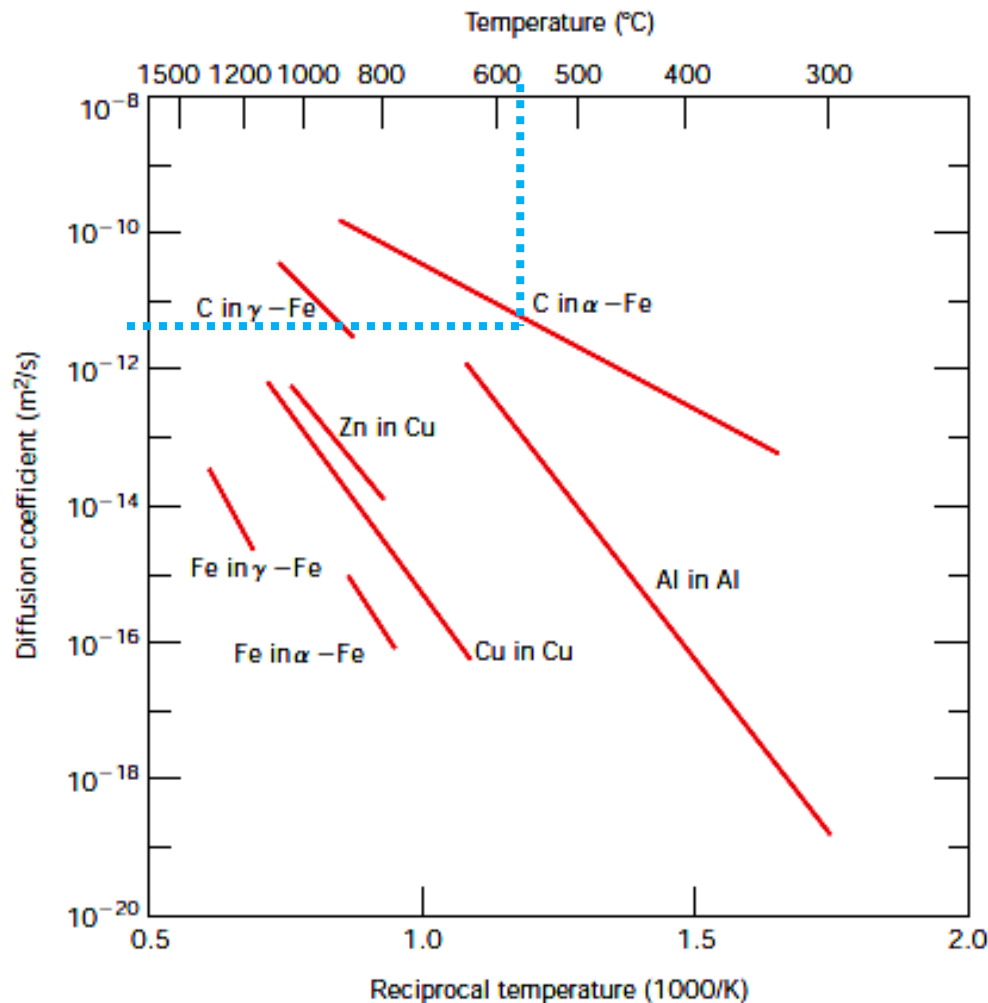
Q_d is activation energy (J/mol or eV/atom)

R is gas constant (8.314 J/mol-K)

T is temperature (K)

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Factors that Influence Diffusion



Plot of the logarithm of the diffusion coefficient versus the reciprocal of absolute temperature for several metals.

$D_{\text{interstitial}} \gg D_{\text{substitutional}}$

$\text{C in } \gamma\text{-Fe}$
 $\text{C in } \alpha\text{-Fe}$

Al in Al
 $\text{Fe in } \gamma\text{-Fe}$
 $\text{Fe in } \alpha\text{-Fe}$

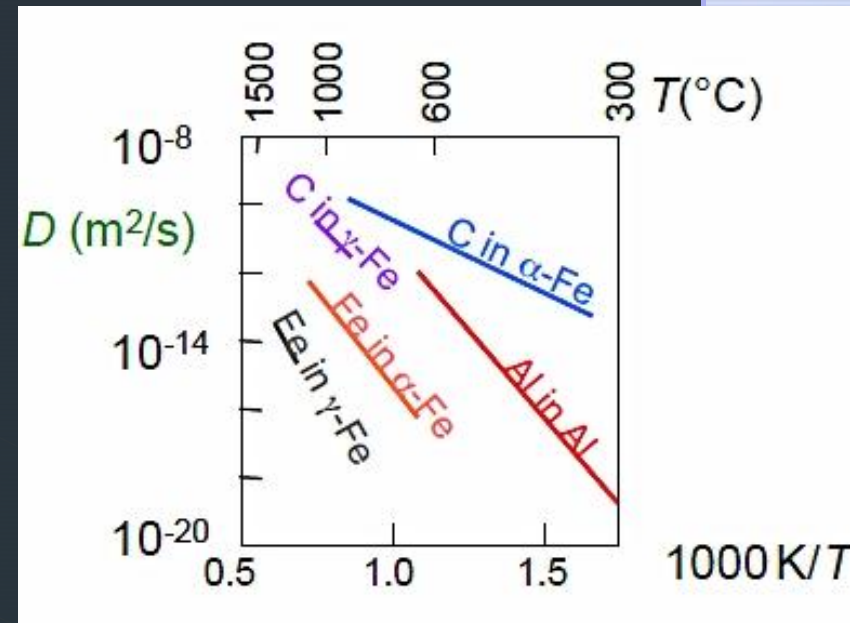
Chapter 4: Diffusion in Solids

Comparing Diffuse in Fe

Is **C** in **FCC Fe** diffusing faster than **C** in **BCC Fe** ?

- **FCC-Fe**: $D_0 = 2.3 \times 10^{-5} (\text{m}^2/\text{s})$
 $Q = 1.53 \text{ eV/atom}$
 $T = 900^\circ\text{C}$, $D = 5.9 \times 10^{-12} (\text{m}^2/\text{s})$

- **BCC-Fe**: $D_0 = 6.2 \times 10^{-7} (\text{m}^2/\text{s})$
 $Q = 0.83 \text{ eV/atom}$
 $T = 900^\circ\text{C}$, $D = 1.7 \times 10^{-10} (\text{m}^2/\text{s})$



- FCC Fe has both higher activation energy Q and D_0 (holes larger in FCC).
- BCC and FCC phase exist over limited range of T (at varying %C).
- Hence, at same T , BCC diffuses faster due to lower Q .

Chapter 4: Diffusion in Solids

Factors that Influence Diffusion

Diffusing Species	Host Metal	$D_0 (m^2/s)$	Activation Energy Q_d		Calculated Values	
			kJ/mol	$eV/atom$	$T(^{\circ}C)$	$D(m^2/s)$
Fe	α -Fe (BCC)	2.8×10^{-4}	251	2.60	500	3.0×10^{-21}
					900	1.8×10^{-15}
Fe	γ -Fe (FCC)	5.0×10^{-5}	284	2.94	900	1.1×10^{-17}
					1100	7.8×10^{-16}
C	α -Fe	6.2×10^{-7}	80	0.83	500	2.4×10^{-12}
					900	1.7×10^{-10}
C	γ -Fe	2.3×10^{-5}	148	1.53	900	5.9×10^{-12}
					1100	5.3×10^{-11}
Cu	Cu	7.8×10^{-5}	211	2.19	500	4.2×10^{-19}
Zn	Cu	2.4×10^{-5}	189	1.96	500	4.0×10^{-18}
Al	Al	2.3×10^{-4}	144	1.49	500	4.2×10^{-14}
Cu	Al	6.5×10^{-5}	136	1.41	500	4.1×10^{-14}
Mg	Al	1.2×10^{-4}	131	1.35	500	1.9×10^{-13}
Cu	Ni	2.7×10^{-5}	256	2.65	500	1.3×10^{-22}

Source: E. A. Brandes and G. B. Brook (Editors), *Smithells Metals Reference Book*, 7th edition, Butterworth-Heinemann, Oxford, 1992.

Chapter 4: Diffusion in Solids

Diffusion and Materials Processing

- ❑ Properties of materials are altered through diffusion
 - **Steelmaking**
 - **Sintering**
 - **Semiconductor doping**
- ❑ **Heat treatment** is used to allow these to occur over a reasonable time frame.

